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CENTER FOR ACADEMIC EXCELLENCE & UN SDG 4 QUALITY EDUCATION

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COMPUTER SCIENCE & AI

Foundations of Machine Learning & Neural Networks

A Rigorous Mathematical and Algorithmic Handbook for Modern Artificial
Intelligence

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Table of Contents

Chapter 1: Mathematical Foundations of Learning Systems	Section 1.0
Chapter 2: Deep Neural Architectures and Training Dynamics	Section 2.0
Chapter 3: Ethical Computing, Algorithmic Bias & SDG 4 Alignment	Section 3.0

This textbook has been compiled by university professors and pedagogical researchers to support deep conceptual comprehension and self-paced mastery. Use the structured review prompts at the end of each module.

CHAPTER 1

Mathematical Foundations of Learning Systems

1.1 Linear Algebra, Matrices, and Vector Topologies

Machine learning fundamentally views datasets as geometric points in high-dimensional vector spaces. A dataset with N samples and D features forms a design matrix X in $\mathbb{R}^{(N \times D)}$. We analyze vector norms, inner products, and orthonormal bases to quantify distances and projections. Spectral theorem and Singular Value Decomposition (SVD) enable low-rank approximations essential for dimensionality reduction and feature extraction.

Key Conceptual Takeaways:

- Vector spaces and inner products define the metric of feature similarity.
- Orthogonal projections minimize mean squared reconstruction errors in Principal Component Analysis (PCA).
- Eigenvalues determine the principal variance axes across complex data manifolds.

1.2 Multivariate Calculus and Gradient Descent Dynamics

Learning parameters θ are updated by navigating high-dimensional loss surfaces $J(\theta)$. For a differentiable loss, the gradient points in the direction of greatest rate of increase. Gradient descent updates parameters in the negative gradient direction: $\theta_{t+1} = \theta_t - \eta * \text{grad } J(\theta_t)$. We study stochastic gradient descent (SGD), momentum acceleration, Adam (adaptive moment estimation), and the role of learning rate scheduling.

Key Conceptual Takeaways:

- Gradient vector contains all partial derivatives of the scalar loss with respect to parameters.
- Second-order Hessian matrix provides curvature information to diagnose saddle points and ill-conditioning.
- Adaptive optimizers dynamically scale updates per coordinate to accelerate convergence.

CHAPTER 2

Deep Neural Architectures and Training Dynamics

2.1 Multi-Layer Perceptrons and the Universal Approximation Theorem

Nonlinear activation functions (ReLU, GELU, LeakyReLU) break linearity and allow deep networks to model arbitrarily intricate decision boundaries. The Universal Approximation Theorem demonstrates that a feedforward network with a single hidden layer and non-polynomial activation can approximate any continuous function on compact subsets of $\mathbb{R}^n$.

Key Conceptual Takeaways:

- Linear transformations without activations collapse any depth of layers into a single linear map.
- Backpropagation computes analytical gradients via recursive application of the chain rule from output to input.
- Vanishing and exploding gradients are mitigated using Residual Connections and Batch/Layer Normalization.

2.2 Convolutional Networks and Sequence Modeling

Spatial invariance in computer vision is modeled through parameter-sharing convolution kernels and spatial pooling. In sequential data, Recurrent Neural Networks and Self-Attention Transformers (Vaswani et al.) calculate attention weights between tokens: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$, revolutionizing modern language understanding.

Key Conceptual Takeaways:

- Convolutions exploit local stationarity and translation equivariance in grid-structured inputs.
- Self-attention computes pairwise affinities across all sequence positions simultaneously in $O(1)$ sequential operations.

CHAPTER 3

Ethical Computing, Algorithmic Bias & SDG 4 Alignment

3.1 Fairness Metrics and Societal Implications of AI Systems

As machine learning informs admissions, grading, and pedagogical recommendations, fairness definitions such as Demographic Parity, Equalized Odds, and Predictive Parity must be rigorously tested. Training data imbalances invariably encode historical prejudice unless active counter-bias preprocessing and adversarial debiasing techniques are integrated.

Key Conceptual Takeaways:

- Demographic parity requires equal positive classification rates across sensitive demographic groups.
- Transparent model explainability (SHAP, LIME) is mandatory for high-stakes educational decision systems.
- AI in Quality Education (SDG 4) must broaden access rather than reinforce existing inequities.

Selected Bibliography & Recommended Literature

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